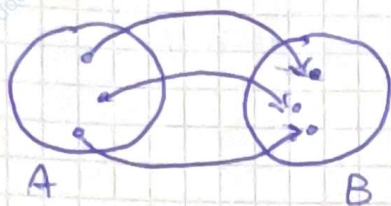


Funzione

Siano A, B insiemi



Defn. una funzione da A a B è una mappa / legge / relazione che associa ad ogni elemento di A uno e un solo elemento di B

scriviamo

$$f: A \rightarrow B$$

dominio codominio

Funzioni Reali:

$$f: A \subseteq \mathbb{R} \rightarrow \mathbb{R}$$

Immagine di A tramite f .

$$f(A) \subseteq \mathbb{R}$$

es.

i. $f(x) = 2x + 3$

$$D_f = \mathbb{R}$$

ii. $f(x) = \sqrt{x}$

$$D: [0, +\infty)$$

iii. $f(x) = \frac{1}{x}$

$$D: \mathbb{R} \setminus \{0\}$$

iv. $f(x) = \tan x$

$$D: \left(-\frac{\pi}{2} + 2k\pi; \frac{\pi}{2} + 2k\pi\right), k \in \mathbb{Z}$$

$$f(A) = \{y \in \mathbb{R} \mid \text{esiste } x \in A \text{ con } f(x) = y\}$$

Defn. Grafico

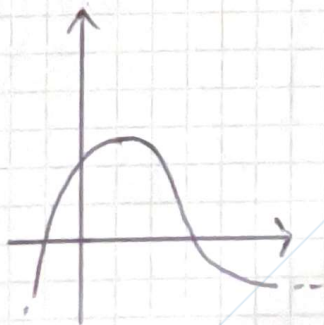
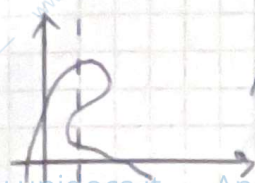


immagine: punti che sono raggiunti dal grafico

$$G(f) = \{(x, y) \in \mathbb{R}^2 \mid x \in \mathbb{R}, y = f(x)\}$$

" $\mathbb{R} \times \mathbb{R}$



NB: NON È FUNZIONE

Gratolo di Funzioni elementari

1) Funzione lineare

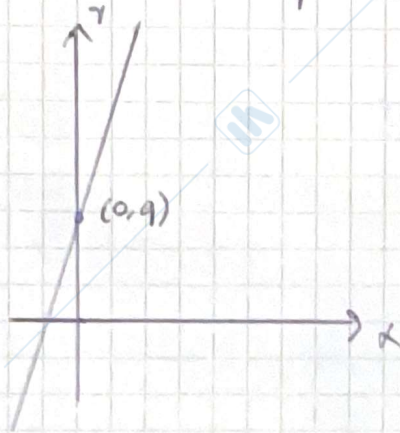
$$m \neq 0, q \in \mathbb{R}$$

$$f(x) = m x + q$$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

pendenza

intersezione
con asse y

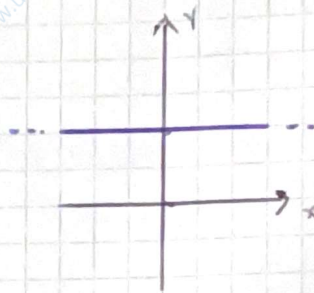


$$\bullet f(0) = q$$

$$\bullet f(1) = m + q$$

2) $m=0$ (nel caso 1)

$$f(x) = q$$



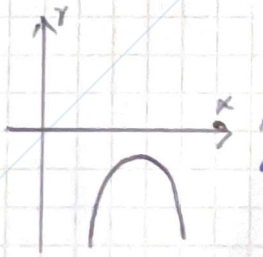
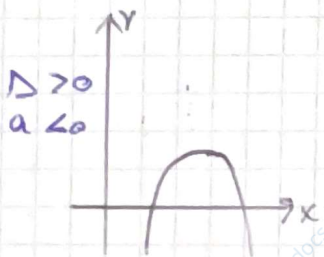
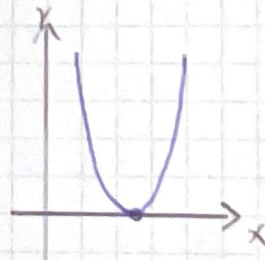
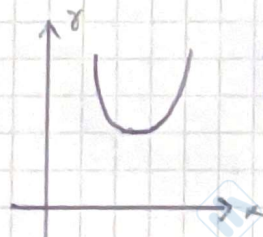
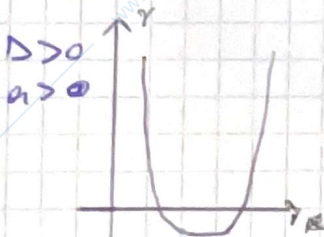
3) Funzione Quadratica

$$a, b, c \neq 0 \in \mathbb{R}$$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = ax^2 + bx + c$$

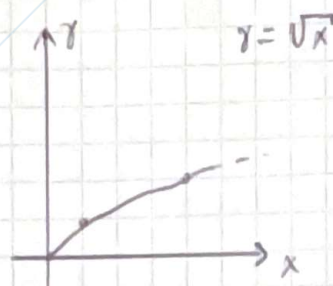
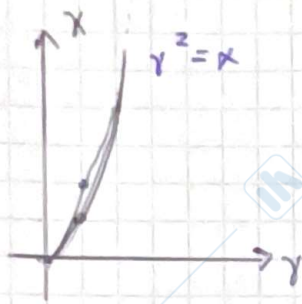
parabole $\begin{cases} \text{verso l'alto} & a > 0 \\ \text{verso il basso} & a < 0 \end{cases}$



4) Funzione radice quadrata

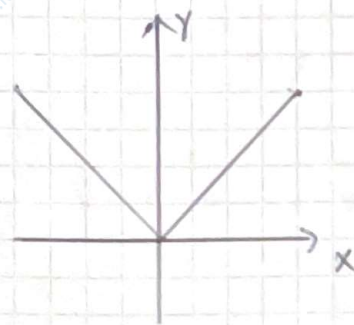
$$f(x) = \sqrt{x} \quad f: [0, +\infty) \rightarrow [0, +\infty)$$

$$(x, y) \in G(f) \Leftrightarrow x \in [0, +\infty) \text{ e } y \in [0, +\infty) \Leftrightarrow \underline{y^2 = x}$$



5) Funzione Modulo

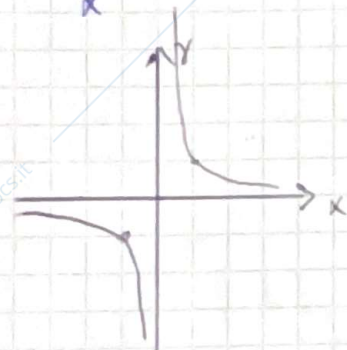
$$f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = |x|$$



$$\begin{cases} x & \text{se } x \geq 0 \\ -x & \text{se } x < 0 \end{cases}$$

6) Funzione Iperbole

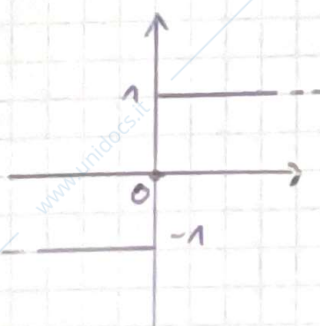
$$f(x) = \frac{1}{x} \quad f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \quad (\mathbb{R} \setminus \{0\})$$



7) Funzione segno

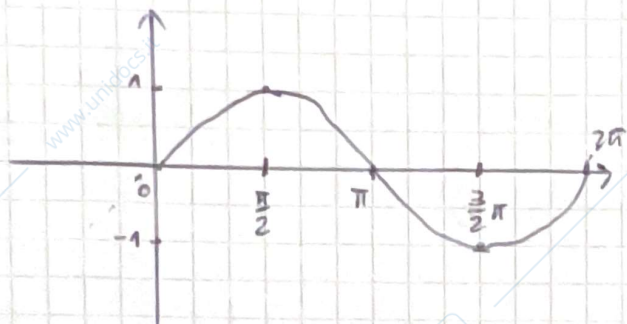
$$f(x) = \text{segn}(x) \\ \text{signum} \\ \text{sign}$$

$$f: \mathbb{R} \rightarrow \{0, -1, 1\}$$



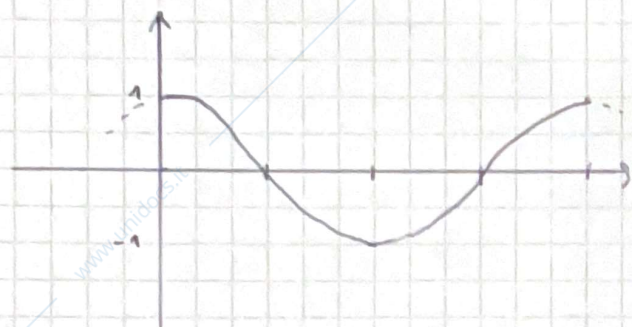
8) $f(x) = \sin x$

$f: \mathbb{R} \rightarrow [-1, 1]$



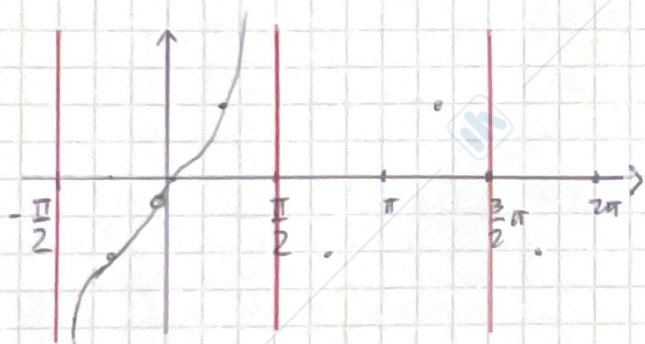
9) $f(x) = \cos x$

$f: \mathbb{R} \rightarrow [-1, 1]$



10) $g(x) = \tan(x)$

$f: \mathbb{R} \setminus \{\frac{\pi}{2} + k\pi\} \rightarrow \mathbb{R}, k \in \mathbb{Z}$



11) altre funzioni:

• parte intera

$f(x) = [x]$

$f: \mathbb{R} \rightarrow \mathbb{Z}$

$= \max \{ k \in \mathbb{Z} \mid k \leq x \}$

• floor arrotondamento verso il basso

• ceil arrotondamento verso l'alto

• mantissa $\{x\} = x - [x]$

Operazioni con le funzioni

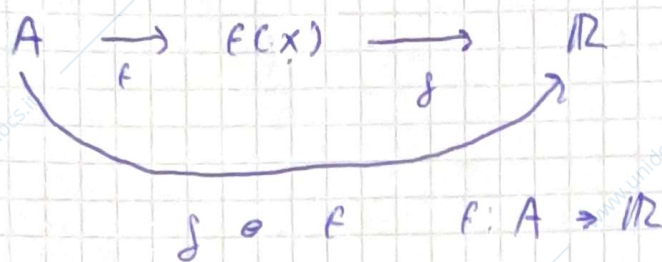
dato $f, g : A \subseteq \mathbb{R} \rightarrow \mathbb{R}$

1) $(f+g)(x) = f(x) + g(x)$

2) $(f \cdot g)(x) = f(x) \cdot g(x)$

3) $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \quad \frac{f}{g} : A \setminus \{x \in A \mid g(x) = 0\} \rightarrow \mathbb{R}$

4) Composizioni di funzioni



$g \circ f(x) = g(f(x))$

es.

$f(x) = \frac{1}{x-5}$

$g(x) = \frac{1}{\sqrt{x}}$

? $g \circ f, f \circ g$

$D_f : \mathbb{R} - \{5\}$

$D_g : (0, +\infty)$

$\bullet (f \circ g)(x) = \frac{\sqrt{x}}{1-5\sqrt{x}} \quad \text{c.e. } \begin{matrix} x \geq 0 \\ x \neq \frac{1}{25} \\ x \neq 0 \end{matrix}$

$\bullet (g \circ f)(x) = -\frac{1}{\sqrt{\frac{1}{x-5}}} \quad D = (5, +\infty)$

Funzione Identità

$I : \mathbb{R} \rightarrow \mathbb{R}$

$I(x) = x$

elemento neutro operazione
composizione

dato f , una funzione generica definita in $\mathbb{R}$:

$f \circ I = I \circ f = f$

oss

o - non è commutativa

$g \circ f \neq f \circ g$

Traslazioni

dato $c \in \mathbb{R}$, $c \neq 0$ e $g(x) = x + c$

1) $(f \circ g)(x) = f(x + c)$

$G(f \circ g)$ si ottiene da $G(f) \begin{cases} a dx & \text{se } x < 0 \\ a dx & \text{se } x > 0 \end{cases}$

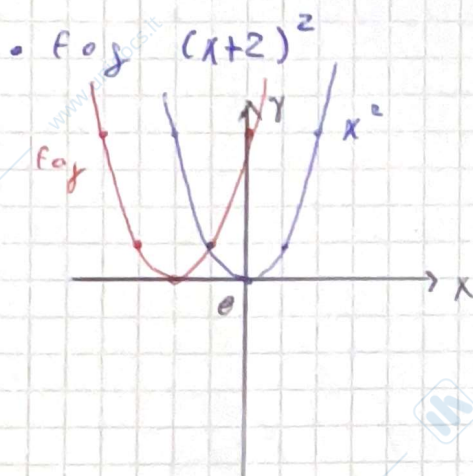
2) $(g \circ f)(x) = f(x) + c$

$G(g \circ f)$ si ottiene da $G(f) \begin{cases} \text{in alto} & c > 0 \\ \text{in basso} & c < 0 \end{cases}$

esempio

$f(x) = x^2$

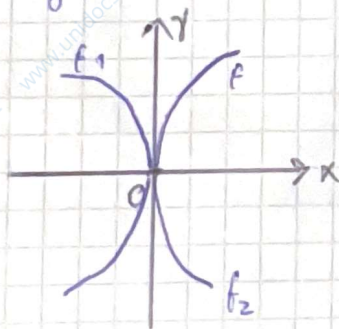
$g(x) = x + 2$



Simmetria

sia $g(x) = -x$

(i) $(f \circ g)(x) = f(g(x)) = f(-x) = f_1(x)$



(ii) $(g \circ f)(x) = g(f(x)) = -f(x) = f_2(x)$

(iii) $(g \circ f \circ g)(x) = -f(-x) = f_3(x)$

Pari, dispari, né pari né dispari.

Sia $A \subseteq \mathbb{R}$ simmetrica

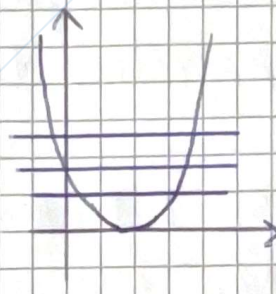
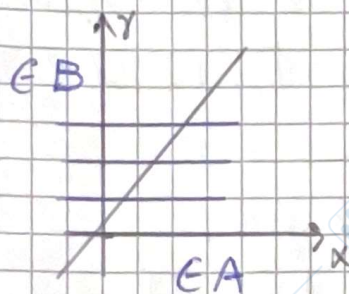
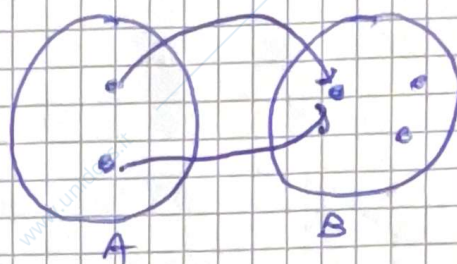
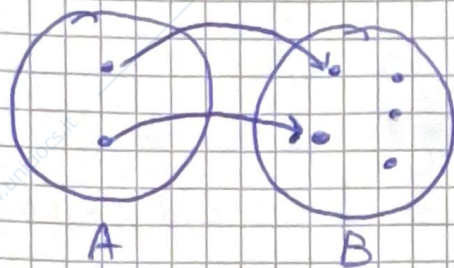
$$x \in A \Leftrightarrow -x \in A$$

Sia $f: A \rightarrow \mathbb{R}$

- è detta pari se $f(-x) = f(x)$

- è detta dispari se $f(-x) = -f(x)$

Funzione iniettiva



Defn Sia $f: A \subseteq \mathbb{R} \rightarrow \mathbb{R}$, è detta iniettiva se ad ogni due valori distinti di A associa due valori distinti di $\mathbb{R}$

- $\forall x \neq z \in A$ si ha $f(x) \neq f(z)$

oppure

- $f(x) = f(z)$, $x, z \in A \Rightarrow x = z$

Funzione monotona

Sia $f: A \subseteq \mathbb{R} \rightarrow \mathbb{R}$ è detta

- (i) monotona strettamente crescente se $\forall x < z$ si ha $f(x) < f(z)$
- (ii) monotona non decrescente se $\forall x \leq z$ $f(x) \leq f(z)$
- (iii) monotona strettamente decrescente se $\forall x < z$ $f(x) > f(z)$
- (iv) monotona non crescente se $\forall x \leq z$ $f(x) \geq f(z)$

Relazione tra monotonia e iniettività

Teo $f: A \subseteq \mathbb{R} \rightarrow \mathbb{R}$

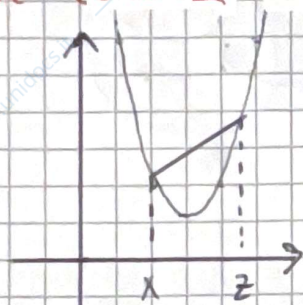
Se f è strett. monotona su $A \Rightarrow f$ è iniettiva
dim.

Per semplicità, supponiamo $f \nearrow$ strettamente

$$\begin{aligned} x \neq z &\Rightarrow x < z \Rightarrow f(x) < f(z) \Rightarrow f(x) \neq f(z) \\ &\quad x > z \Rightarrow f(x) > f(z) \end{aligned}$$

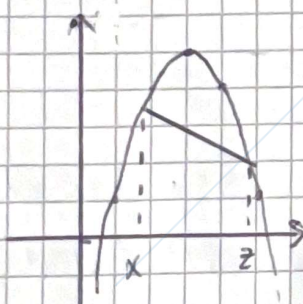
oss. NON VALE $\Leftarrow$ se induttiva

Funzione convessa



Sia $f: I \subseteq \mathbb{R} \rightarrow \mathbb{R}$, I intervallo.
Allora f è convessa se $\forall x, z \in I, x < z$
si ha il segmento che unisce $(x, f(x))$ e
 $(z, f(z))$ si trova sopra il grafico di f in $[x, z]$

Funzione concava



Sia $f: I \subseteq \mathbb{R} \rightarrow \mathbb{R}$, I intervallo.
Allora f è concava se $\forall x, z \in I, x < z$
si ha il segmento che unisce $(x, f(x))$ e $(z, f(z))$
si trova sotto il grafico della funzione in $[x, z]$

Segno e zeri di una funzione data $f: A \subseteq \mathbb{R} \rightarrow \mathbb{R}$

(i) f si dice positiva su A se $f(x) > 0$

f si dice non negativa su A se $f(x) \geq 0$

f si dice negativa su A se $f(x) < 0$

f si dice non positiva su A se $f(x) \leq 0$

(ii) x si chiama zero della funzione se $f(x) = 0$



Ripasso prodotto - composizione

Il prodotto in $\mathbb{R}$

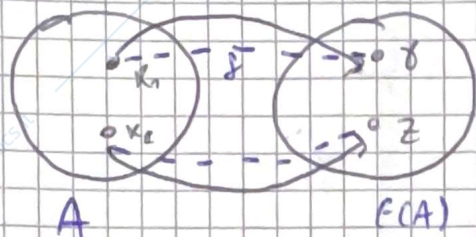
- è commutativa
- è associativa
- $\exists$ elemento neutro (1)
- $\exists$ elemento inverso ($x \cdot y = 1$)

Composizione di funzione (o)

- non è commutativa
- è associativa
- $\exists$ elemento neutro $I(x) = x$
- $f \circ g = I$

Funzione Inversa

$f: A \subseteq \mathbb{R} \rightarrow f(A)$ è f. iniettiva



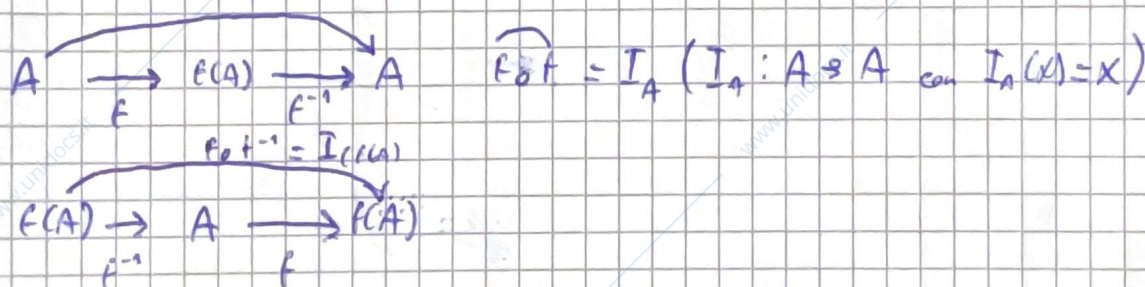
Allora, $\forall y \in f(A) \exists! x \in A \mid f(x) = y$ $\left(\begin{array}{l} f(1) = 5 \\ \Rightarrow g(5) = 1 \end{array} \right)$

defn $g: f(A) \rightarrow A \mid g(y) = x$

g è funzione detta inversa e usiamo la notazione $f^{-1} = g$

Proprietà

Se $f: A \rightarrow f(A)$ iniettiva, $f^{-1}: f(A) \rightarrow A$ funz. inversa



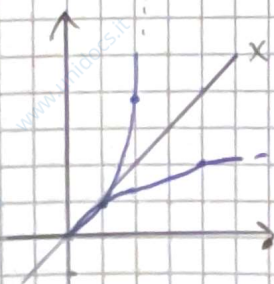
Esemp

1) $f(x) = 2x + 1$ $f: \mathbb{R} \rightarrow \mathbb{R}$

calcolo funz. inversa

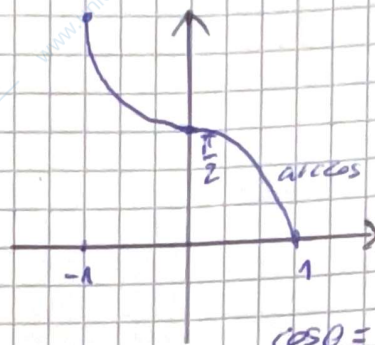
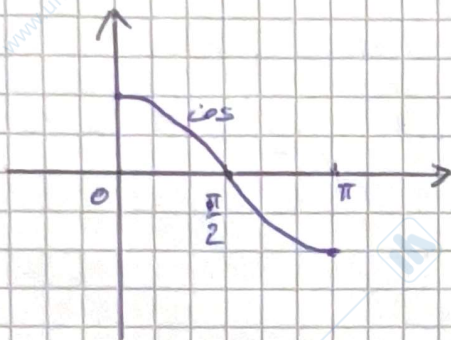
$$f(x) = y \quad y = 2x + 1 \quad \frac{y-1}{2} = x \quad f^{-1}(x) = \frac{x-1}{2}$$

2) $f(x) = x^2$ $f: [0; +\infty) \rightarrow [0; +\infty)$



Funzioni goniometriche inverse

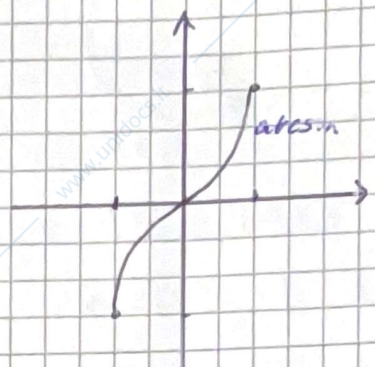
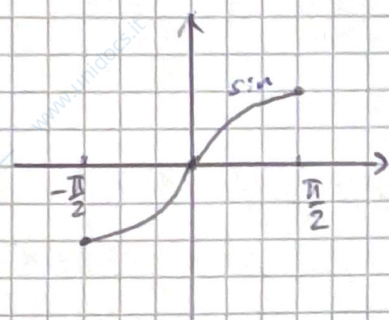
1) $f(x) = \cos x$ $f: [0, \pi] \rightarrow [-1, 1]$



$\cos 0 = 1 \Rightarrow \arccos 1 = 0$

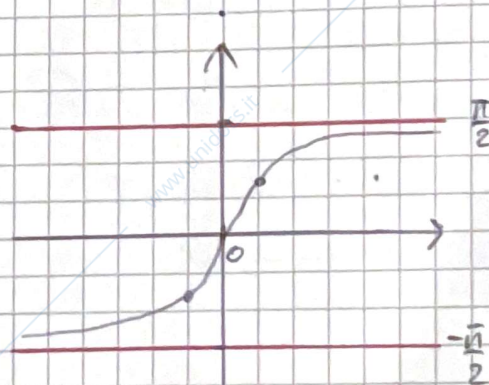
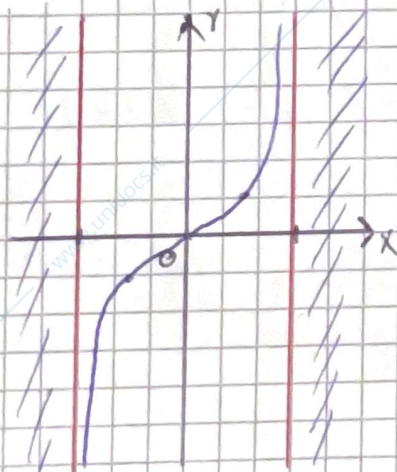
$f([0, \pi]) = [-1, 1]$ • f. iniettiva $\Rightarrow \exists$ inversa: $\arccos: [-1, 1] \rightarrow [0, \pi]$

2) $f(x) = \sin x$ $f: [-\frac{\pi}{2}, \frac{\pi}{2}] \rightarrow [-1, 1]$



$f([-\frac{\pi}{2}, \frac{\pi}{2}]) = [-1, 1]$ • f. iniettiva $\Rightarrow \exists$ inversa: $\arcsin: [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$

3) $f(x) = \tan x$ $f: (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}$



$f(-\frac{\pi}{2}, \frac{\pi}{2}) = \mathbb{R}$ • iniettiva $\Rightarrow \exists$ inversa: $\arctan: \mathbb{R} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$

Funzione Limitata

Sia $f: A \subseteq \mathbb{R} \rightarrow \mathbb{R}$

1) f è limitata superiormente se $f(A)$ è un insieme superiormente limitato oppure se $\exists M \in \mathbb{R} \mid f(x) \leq M \quad \forall x \in A$

2) f è limitata inferiormente se $f(A)$ è un insieme inferiormente limitato oppure se $\exists m \in \mathbb{R} \mid f(x) \geq m \quad \forall x \in A$

es.

$$f(x) = \arctan x$$

$$f(\mathbb{R}) = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\inf f = -\frac{\pi}{2}$$

$$\sup f = \frac{\pi}{2}$$

Max Globale

$f: A \subseteq \mathbb{R} \rightarrow \mathbb{R}$, $x^* \in A$ è detto punto di massimo globale se $f(x^*) \geq f(x) \quad \forall x \in A$

Min Globale

$f: A \subseteq \mathbb{R} \rightarrow \mathbb{R}$, $x_* \in A$ è detto punto di minimo globale se $f(x_*) \leq f(x) \quad \forall x \in A$

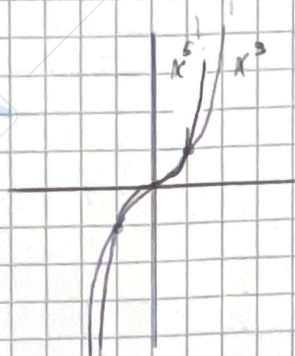
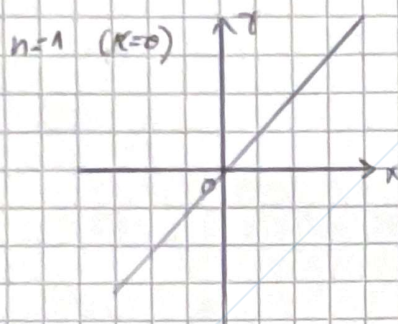
Grafici di funzioni elementari

1) Funzioni potenza a esp. naturale

$$f(x) = x^n \quad n \in \mathbb{N}$$

(i) n dispari

$$f(x) = x^{2k+1} \quad k \in \mathbb{N}_0 \quad f: \mathbb{R} \rightarrow \mathbb{R}$$



$$x^3 \leq x^5$$

$$x > 0 \iff x \geq 1$$

$$x < 0 \iff x \leq -1$$

• funz. dispari $f(-x) = -f(x)$

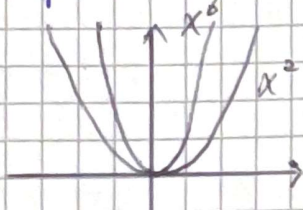
• funz. strettamente crescente

• f. iniettiva in $\mathbb{R}$

$$\exists \text{ f. inverse } x = \sqrt[n]{y}$$

(ii) n pari $f(x) = x^{2k}$ $k \in \mathbb{N}$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$



• f. pari

• f. strett. $\nearrow$ $[0, +\infty)$

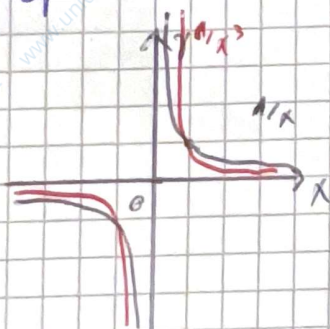
• f. strett. $\searrow$ $(-\infty, 0]$

2 Funz. potenza a intero negativo

$$f(x) = \frac{1}{x^n} \quad \text{opp.} \quad x^{-n} \quad n \in \mathbb{N}$$

$$f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$$

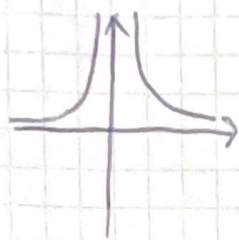
(i) dispari $f(x) = \frac{1}{x^{2k+1}}$ $k \in \mathbb{N}_0$



$$\frac{1}{x} \leq \frac{1}{x^3} \quad : x > 0$$

$$x \leq 1$$

(ii) n pari: $f(x) = x^{-2k}$ $f(x) = \frac{1}{x^{2k}}$ $k \in \mathbb{N}$ $f: \mathbb{R} \setminus \{0\} \rightarrow (0, +\infty)$

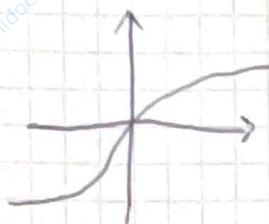


o pari, non invertibile

3) Funz. potenza a esponente $1/n$

$f(x) = x^{\frac{1}{n}}$ $n \in \mathbb{N}$ (funz. inversa di x^n) $f^{-1}(x)$

(i) n dispari $x^{\frac{1}{2k+1}}$ $k \in \mathbb{N}_0$ $f: \mathbb{R} \rightarrow \mathbb{R}$ $x^{\frac{1}{n}} = \begin{cases} \sqrt[n]{x} & x \geq 0 \\ -\sqrt[n]{-x} & x < 0 \end{cases}$

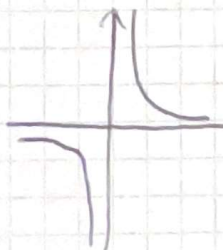


(ii) n pari $x^{\frac{1}{2k}}$ $k \in \mathbb{N}_0$ ~~$f: \mathbb{R} \rightarrow \mathbb{R}$~~
 $f: [0, +\infty) \rightarrow [0, +\infty)$



4) Funz. potenza a esponente $-1/n$

(i) n dispari $f(x) = x^{-\frac{1}{2k+1}}$ $k \in \mathbb{N}_0$



5) Funz. potenze a esp m/n

$$x^{\frac{m}{n}} = \sqrt[n]{x^m} = (\sqrt[n]{x})^m$$

1) $\frac{m}{n} > 1$ • m dispari, n pari

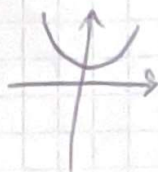
$$f: [0, +\infty) \rightarrow [0, +\infty)$$

$$x^{\frac{3}{2}}$$



• m pari, n dispari

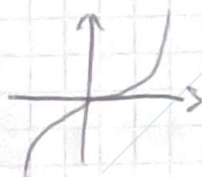
$$x^{\frac{4}{3}}$$



$$f: \mathbb{R} \rightarrow [0, +\infty)$$

• m, n dispari

$$x^{\frac{5}{3}}$$



$$f: \mathbb{R} \rightarrow \mathbb{R}$$

2) $0 < \frac{m}{n} < 1$

• m dispari, n pari

$$x^{\frac{3}{4}}$$



$$f: [0, +\infty) \rightarrow [0, +\infty)$$

• n pari, m dispari

$$x^{\frac{2}{3}}$$



$$f: \mathbb{R} \rightarrow [0, +\infty)$$

• m, n dispari

$$x^{\frac{5}{3}}$$



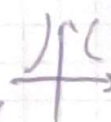
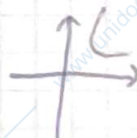
3) $\frac{n}{m} < 0$

$$x^{-\frac{1}{2}}$$

$$f: (0, +\infty) \rightarrow (0, +\infty)$$

$$f: \mathbb{R} \rightarrow (0, +\infty)$$

$$f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$$



c) Funz. esp. reale

$$f(x) = x^\alpha$$

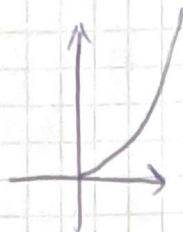
$$f: [0, +\infty) \rightarrow [0, +\infty) \quad \alpha > 0$$

$$f: (0, +\infty) \rightarrow (0, +\infty) \quad \alpha < 0$$

$$0 < \alpha < 1$$



$$\alpha > 1$$



$$\alpha < 0$$



$$x^a \cdot x^b$$

$$\frac{x^a}{x^b}$$

$$(x^a)^b$$

$$\text{es. } x^{\frac{2}{3}} \leq 2 - x^2 \quad \text{graficamente}$$

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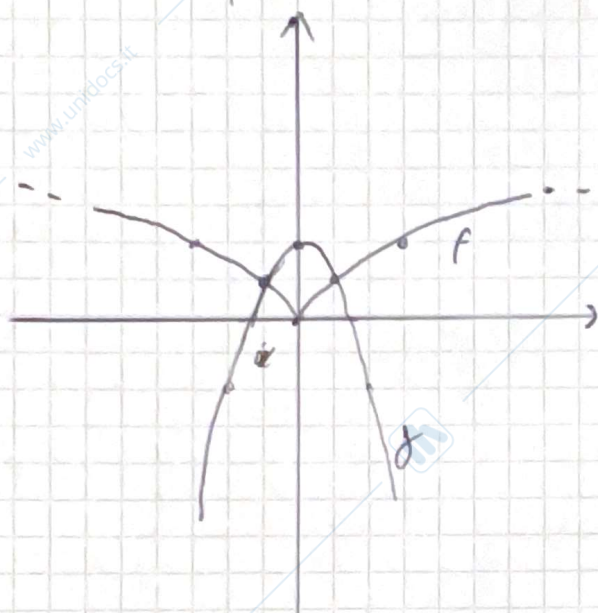
Alcune disegnature da risolvere con gli assi

1) $x^{2/3} \leq 2 - x^2$

$f(x) = x^{2/3} = \sqrt[3]{x^2}$
 $g(x) = 2 - x^2$

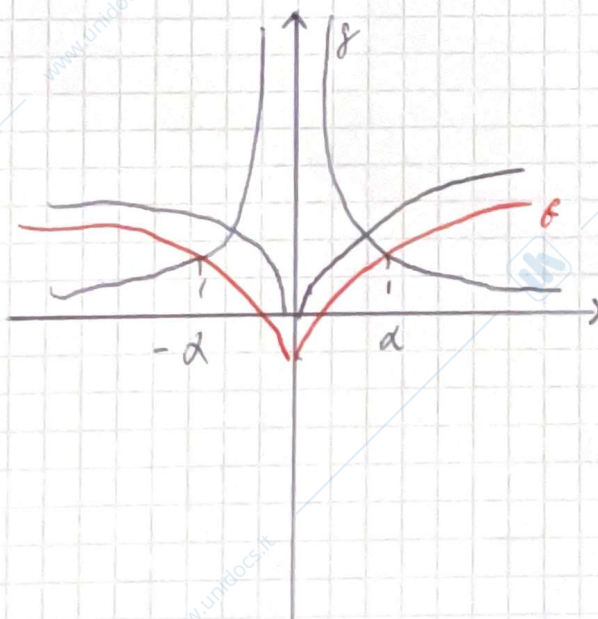
$f: \mathbb{R} \rightarrow \mathbb{R}$

$g: \mathbb{R} \rightarrow \mathbb{R}$



$S = [-1, 1]$

2) $\sqrt{|x|} - 1 \geq x^{-2/3}$



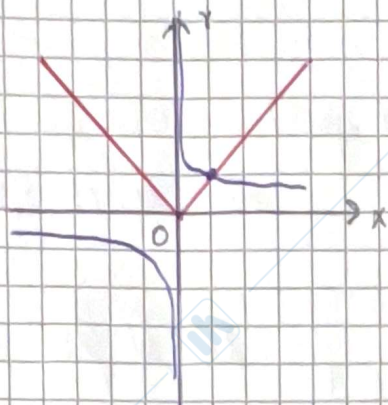
$f(x) = \sqrt{|x|} \rightarrow \sqrt{x} \text{ se } x \geq 0$

$g(x) = \frac{1}{\sqrt[3]{x^2}}$

$D_f: \mathbb{R} \setminus \{0\}$

$S = (-\infty, -1] \cup [1, +\infty)$

$$\exists) |x| > x^{-\frac{1}{5}}$$



$$S: (-\infty, 0) \cup (1, +\infty)$$

Ricorrenza

$$a \leq b \quad a, b \in \mathbb{R}$$

$$= a^{2k+1} \leq b^{2k+1} \quad \forall k \in \mathbb{N}$$

$$= 0 \leq a \leq b \quad a^{2k} \leq b^{2k}$$

$$= a \leq b < 0 \quad a^{2k} \geq b^{2k}$$